Sr. No.			Description	UOM (Wherever Applicable)	Data (Common For All Models)
A			General Points		
	1		Cooling Capacity	ton_R	Refer KCPL Chiller Selection System Software
	2		Power Consumption	kW	Refer KCPL Chiller Selection System Software
	3		Specific Power Consumption	kW/ton_R	Refer KCPL Chiller Selection System Software
	4		Co-Efficient of Performance (COP)	kW/kW	Refer KCPL Chiller Selection System Software
	5		No. of Compressors	Nos.	Contact To Engineering Team
	6		No. of Individual Refrigerant Circuits	Nos.	Contact To Engineering Team
	7		Refrigerant		
		i	Name	-	R134a
		ii	Quantity	kg	Refer KCPL Chiller Selection System Software
		iii	Technical Specifications	-	Refer ESP-18-19-003
	8		Sound Pressure Level		
		i	Noise Level	dB	Refer ESP-18-19-001
		ii	Measuring Standard	-	ANSI/AHRI Standard 575-2008
	9		Insulation Details		
		i	Material	-	Closed Cell Nitrile Foam
		ii	Insulation Thickness on Various Parts	-	For Standard Temperature Range (LWT upto -10 ^{0}C)
			Evaporator Shell	mm	32
			Evaporator Tubesheet	mm	19
			Evaporator Dished End	mm	19
			Evaporator M.W.Box (If Applicable)	mm	19
			Evaporator Support Plate	mm	19
			Compressor Motor Body	mm	19
			Suction Line Assembly	mm	19
			Liquid Line Assembly	mm	9
		iii	Insulation Thickness on Various Parts	-	For Brine Temperature Range (LWT below -10 ^{0}C)
			Evaporator Shell	mm	51 (32+19)
			Evaporator Tubesheet	mm	32
			Evaporator Dished End	mm	32
			Evaporator M.W.Box (If Applicable)	mm	32
			Evaporator Support Plate	mm	32
			Compressor Motor Body	mm	28 (19+9)
			Suction Line Assembly	mm	28 (19+9)
			Liquid Line Assembly	mm	19
		iv	Density	kg/m^3	76.6
		v	Thermal Conductivity	W/m.K	0.035 (at 0 ^{0}C Mean Temperature)
		vi	Standard	-	IS 14164
		vii	Adhesive	-	Blend of Synthetic Polymers and Synthetic Resin
		viii	Insulation Specifications	-	Refer ESP-18-19-004
	10		Vibration		
		i	Vibration Level	mm/sec	Less than 1.5 mm/sec
		ii	Vibration control	-	Rubber Pads (Standard) / Spring Isolators (At an Additional Cost)
		iii	Standard	-	IS 12075
	11		Painting Specification		
		i	Paint Type	-	RAL 7035
		ii	Standard	-	Coating as per KCPL Standards
	12		Overall Dimensions		
		i	Approx. Length	mm	Refer KCPL Chiller Selection System Software
		ii	Approx. Width	mm	Refer KCPL Chiller Selection System Software
		iii	Approx. Height	mm	Refer KCPL Chiller Selection System Software
	13		Space Clearances Required		
		i	Plain End Side (For Tube Cleaning)	mm	Contact To Engineering Team
		ii	All Other Sides	mm	Contact To Engineering Team
		iii	Overhead	mm	Contact To Engineering Team
	14		Weight		
		i	Approx. Shipping Weight	kg	Refer KCPL Chiller Selection System Software
		ii	Approx. Operating Weight	kg	Refer KCPL Chiller Selection System Software
	15		Cable Sizes		
		i	Aluminum Cable	-	Refer ESP-14-15-01
		ii	Copper Cable	-	Refer ESP-14-15-01
B			Compressor Details		
	1		Make	-	Kirloskar Chillers Private Limited
	2		Type / Description	-	Semi-Hermetic Twin Screw Compressor
	3		Model	-	Refer KCPL Chiller Selection System Software
	4		Drive	-	Direct Driven by Rotor Shaft
	5		Capacity Control Percentage	-	Contact To Engineering Team
	6		Type of Capacity Control	-	Stepless
	7		Capacity Control Mechanism	-	Slide Valve Mechanism
	8		Volumetric Ratio	-	Fixed Ratio (2.2)

Sr. No.			Description	UOM (Wherever Applicable)	Data (Common For All Models)
	9		Design and Test Parameters		
		i	Design Pressure	bar	30
		ii	Test Pressure (Pneumatic)	bar	33
		iii	Design Temperature	^{0}C	120
		iv	Max. Allowable Discharge Temperature	^{0}C	120
	10		Bearings		
		i	Types of Bearings	-	Roller Bearings - For Radial Load
					Angular Contact Roller Bearing - For Axial Load
		ii	Material of Construction	-	Steel
		iii	Life of Bearing	Hours	50000
		iv	Class of Bearing	-	Proprietary Data
	11		Lubrication		
		i	Type	-	Lubrication by Differential Pressure Mechanism
		ii	Lubricating Oil	-	Synthetic Oil
		iii	Grade of Lubricating Oil	-	Proprietary Data
		iv	Quantity	Liter	Refer KCPL Chiller Selection System Software
	12		Compressor Components MOC		
		i	Screw	-	Alloy Steel
		ii	Casing	-	Cast Iron
		iii	Shaft	-	Alloy Steel
		iv	Rotor	-	Aluminum Alloy
	13		Physical Data of Compressor		
		i	Screw Construction	-	Twin Screw
		ii	No. of Lobes Male Rotor	Nos.	5
		iii	No. of Lobes Female Rotor	Nos.	6
		iv	Male Rotor Diameter (mm)	mm	Proprietary Data
		v	Female Rotor Diameter (mm)	mm	Proprietary Data
		vi	Driving Rotor	-	Male Rotor
	14		Oil Filter		
		i	Micron Rating	Micron	10
		ii	Material of Construction	-	Resin Impregnated Fibres
		iii	Quantity	Nos.	1 No. per Compressor
	15		Copressor Isolation Type		
		i	At Suction	-	No Isolation
		ii	At Discharge	-	Shut-off Valve
C			Compressor Motor Details		
	1		Make	-	Kirloskar Approved Vendor
	2		Motor Type	-	Semi-Hermetic Squirrel Cage Induction Motor
	3		Type of Duty	-	Continuous
	4		Motor Rating	kW	Refer KCPL Chiller Selection System Software
	5		Motor Speed (Synchronous)	RPM	3000
	6		Ingress Protection (IP)	-	NA, Being Semi-Hermetic Type
	7		GD^2 of Rotor	-	Proprietary Data
	8		Whether SPDP or TEFC?	-	NA, Being Semi-Hermetic Type
	9		Power Supply Details (Standard)		
		i	Supply Voltage	V	400
		ii	Permissible Voltage Variation	%	±10%
		iii	Frequency	Hz	50
		iv	Permissible Frequency Variation	%	±3%
		v	Phase	-	3
	10		Performance Indicators		
		i	Motor Efficiency Class	-	NA
		ii	Motor Power	kW	Refer KCPL Chiller Selection System Software
		iii	Motor Efficiency	-	Consult with Engineering Department on Case to Case Basis
		iv	Power Factor	-	Consult with Engineering Department on Case to Case Basis
		v	Class of Insulation	-	Class F
	11		Motor Cooling		
		i	Motor Cooling Type	-	Refrigerant Cooled
		ii	Cooling Mechanism	-	Suction Gas
		iii	Temperature at full load	^{0}C	10 to 15 (At Normal Condtions)
	12		Current Details		
		ii	Rated Load Current	A	Refer KCPL Chiller Selection System Software
		iii	Full Load Current	A	Refer KCPL Chiller Selection System Software
		iv	Inrush/Starting Current	A	Refer KCPL Chiller Selection System Software
		v	Locked Rotor Current	A	Refer KCPL Chiller Selection System Software
		vi	Starting Torque	N.m	Contact To Engineering Team
		vii	No Load Current	A	Contact To Engineering Team
		viii	Acceleration Time to Reach Rated Speed	Sec	2 to 3
	13		Control Settings		
		i	No. of Starts per Hour	Nos.	4

Sr. No.			Description	UOM (Wherever Applicable)	Data (Common For All Models)
		ii	Time Between STOP to START	Sec	300
		iii	Time Between START to START	Sec	900
D			Power Supply (Standard-Chiller Icomer)		
	1		Supply Voltage	V	415
	2		Permissible Voltage Variation	%	±10%
	3		Frequency	Hz	50
	4		Permissible Frequency Variation	%	±3%
	5		Phase	-	3
	6		Control Voltage	V	230 (Standard)
					110 (Special-Optional)
	7		Supply Wire System	-	3 Phase - 4 Wire System (Standard)
					3 Phase - 3 Wire System (Special-Optional)
	8		Fault Level at Busbar	kA	As per KCPL Standard Practice
E			Oil Separator Details		
	1		Type	-	Horizontal Type
	2		Internal Structure	-	Baffle - Demister Arrangement
	3		Method of Oil Separation	-	Separation by "Filtering Effect" Obtained Through Demister
	4		Material of Construction		
		i	Body and Other Parts	-	Mild Steel (Refer "MOC" Sheet)
		ii	Demister	-	SS
	5		Physical Details		
		i	Shell Diameter	inch	Contact To Engineering Team
		ii	Approx. Length	mm	Contact To Engineering Team
	6		Seperation Efficiency	%	99
	7		Oil Heater Details		
		i	Make	-	Kirloskar Approved Vendor
		ii	Quantity	Nos.	Contact To Engineering Team
		iii	Power Supply	V	230
		iv	Rating	W	250
F			Oil Cooler	-	Not Applicable
G			Evaporator Details		
	1		Model	-	Refer KCPL Chiller Selection System Software
	2		Design Code	-	As per KCPL Standards
	3		Type	-	Shell and Tube Flooded Design
	4		Tube Side (Fluid)	-	Chilled Water
	5		Shell Side (Fluid)	-	Refrigerant
	6		Design Parameters		
		i	Design Temperature (Refrigerant Side)	^0C	65
		ii	Max. Operating Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		iii	Design Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		iv	Test pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		v	Testing method (Refrigerant Side)	-	Refer ESP-07-08-107
		vi	No. of Passes (Refrigerant Side)	Nos.	Single Pass
		vii	Design Temperature (Water Side)	^0C	65
		viii	Max. Operating Pressure (Water Side)	bar	Refer ESP-07-08-107
		ix	Design Pressure (Water Side)	bar	Refer ESP-07-08-107
		x	Test pressure (Water Side)	bar	Refer ESP-07-08-107
		xi	Testing method (Water Side)	-	Refer ESP-07-08-107
		xii	No. of Passes (Water Side)	Nos.	Two Pass
		xiii	Water Velocity	m/s	Less than 3 m/s
		xiv	Inlet Pressure (Water Side)	bar	Depends on Site Piping Layout (Maximum Allowable - 9.4 bar)
		xv	Evaporating Temperature	^0C	Consult with Engineering Department on Case to Case Basis
	7		Physical Data of Evaporator		
		i	Overall Length of Evaporator	ft	Contact To Engineering Team
		ii	Shell Diameter	inch	Contact To Engineering Team
		iii	Shell Thickness	mm	Contact To Engineering Team
		iv	Approx. Shell Length	mm	Contact To Engineering Team
		v	Material of Construction of Shell	-	Mild Steel
		vi	Material Standard of Shell	-	Refer "MOC" Sheet
		vii	Tube Type/ Nature of Tube Surface	-	Integral Helical Fins on the Outside Surface and Integral Helical Ridges on the Inside Surface
		viii	Tube Length	mm	Refer "HX Details" Sheet
		ix	Tube Diameter	mm	Refer "HX Details" Sheet
		x	Tube Thickness	mm	Refer "HX Details" Sheet
		xi	Material of Construction of Tube	-	Cu
		xii	Material Standard of Tube	-	Refer "MOC" Sheet
		xiii	Water Volume in Evaporator	Liter	Refer KCPL Chiller Selection System Software
	8		Water Box Details		
		i	Type	-	Standard - Dish Ends (M.W.Box - Optional)
		ii	Material	-	Mild Steel

Sr. No.			Description	UOM (Wherever Applicable)	Data (Common For All Models)
		iii	Material Standard	-	Refer "MOC" Sheet
		iv	Nozzle size	NB	Refer KCPL Chiller Selection System Software
		v	End connection	-	Standard - Victaulic Conn. (Flanged Conn. - Optional)
		vi	MOC of Water Side Gasket	-	NAM AF 120
		vii	MOC of Refrigerant Side Gasket	-	NAM AF 159
	9		Accessories Provided		
		i	Pressure Relief Valve	-	Spring Loaded (For Safety Valve Set Pressure Refer ESP)
		ii	Drain/Vent Valves	Inch	Plugged Connection Provided (3/8" NPT)
H			Condenser Details		
	1		Model	-	Refer KCPL Chiller Selection System Software
	2		Design Code	-	As per KCPL Standards
	3		Type	-	Shell and Tube Flooded Design
	4		Tube Side (Fluid)	-	Chilled Water
	5		Shell Side (Fluid)	-	Refrigerant
	6		Design Parameters		
		i	Design Temperature (Refrigerant Side)	^0C	100
		ii	Max. Operating Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		iii	Design Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		iv	Test pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		v	Testing method (Refrigerant Side)	-	Refer ESP-07-08-107
		vi	No. of Passes (Refrigerant Side)	Nos.	Single Pass
		vii	Design Temperature (Water Side)	^0C	100
		viii	Max. Operating Pressure (Water Side)	bar	Refer ESP-07-08-107
		ix	Design Pressure (Water Side)	bar	Refer ESP-07-08-107
		x	Test pressure (Water Side)	bar	Refer ESP-07-08-107
		xi	Testing method (Water Side)	-	Refer ESP-07-08-107
		xii	No. of Passes (Water Side)	Nos.	Two Pass
		xiii	Water Velocity	m/s	Less than 3 m/s
		xiv	Inlet Pressure	bar	Depends on Site Piping Layout (Maximum Allowable - 9.4 bar)
		xv	Condensing Temperature	^0C	Consult with Engineering Department on Case to Case Basis
	7		Physical Data of Condenser		
		i	Overall Length of Condenser	ft	Contact To Engineering Team
		ii	Shell Diameter	inch	Contact To Engineering Team
		iii	Shell Thickness	mm	Contact To Engineering Team
		iv	Shell Length	mm	Contact To Engineering Team
		v	Material of Construction of Shell	-	Mild Steel
		vi	Material Standard of Shell	-	Refer "MOC" Sheet
		vii	Tube Type/ Nature of Tube Surface	-	Integral Helical Fins on the Outside Surface and Integral Helical Ridges on the Inside Surface
		viii	Tube Length	mm	Refer "HX Details" Sheet
		ix	Tube Diameter	mm	Refer "HX Details" Sheet
		x	Tube Thickness	mm	Refer "HX Details" Sheet
		xi	Material of Construction of Tube	-	Cu
		xii	Material Standard of Tube	-	Refer "MOC" Sheet
		xiii	Water Volume in Condenser	Liter	Refer KCPL Chiller Selection System Software
	8		Water Box Details		
		i	Type	-	Standard - Dish Ends (M.W.Box - Optional)
		ii	Material	-	Mild Steel
		iii	Material Standard	-	Refer "MOC" Sheet
		iv	Nozzle size	NB	Refer KCPL Chiller Selection System Software
		v	End connection	-	Standard - Victaulic Conn. (Flanged Conn. - Optional)
		vi	MOC of Water Side Gasket	-	NAM AF 120
		vii	MOC of Refrigerant Side Gasket	-	NAM AF 159
	9		Accessories Provided		
		i	Pressure Relief Valve	-	Spring Loaded (For Safety Valve Set Pressure Refer ESP)
		ii	Drain/Vent Valves	Inch	Plugged Connection Provided (3/8" NPT)
I			Suction Line		
	1		Design Code	-	ASME B31.3
	2		Isolation Valve	-	No Isolation
	3		Material of Construction	-	Carbon Steel
	4		Material Standard	-	Refer "MOC" Sheet
	5		Angle Valve	-	Provided on Suction Line For Oil Recovery Line
J			Discharge Line		
	1		Design Code	-	ASME B31.3
	2		Isolation Valve	-	Shut-off Valve
	3		Material of Construction	-	Carbon Steel
	4		Material Standard	-	Refer "MOC" Sheet
	5		Skin Type Thermowell	-	Provided on Discharge Line For Discharge Temp. Sensor
K			Liquid Line		
	1		Design Code	-	ASME B31.3

Sr. No.			Description	UOM (Wherever Applicable)	Data (Common For All Models)
	2		Expansion Valve		
		i	Type	-	Electronic Expansion Valve
		ii	Make	-	Refer "Make List" Sheet
		iii	Quantity	Nos.	Contact To Engineering Team
		iv	Sight Glass	-	Inbuilt
		v	Moisture Indicator	-	NA
	3		Filter Drier	-	NA
	4		Material of Construction	-	Copper
	5		Material Standard	-	Refer "MOC" Sheet
L			Desuperheater		
	1		Type	-	Plate Type
	2		Quantity	Nos.	One per Compressor
	3		Operating Conditions		
		i	Heat Duty	kW	Depends on Working Conditions
		ii	Hot Water Inlet Temperaure	^{0}C	Depends on Site Conditions (Max. Possible - 40)
		iii	Hot Water Outlet Temperaure	^{0}C	Max. Possible - 45
		iv	Hot Water Flow Rate	L/s	Depends on Working Conditions
	4		Material of Construction	-	Brazzed PHE, Plate Material - SS
	5		Water Side End connection Details		
			Water Inlet Connection	NB	Consult with Engineering Department on Case to Case Basis
			Water Outlet Connection	NB	Consult with Engineering Department on Case to Case Basis
	6		Pressure Drop		
		i	Water Side	bar	less than 0.5
		ii	Refrigerant Side	bar	Proprietary Data
M			Economizer	-	Not Applicable
N			Starter and Control Panel		
	1		Panel Enclosure	-	Starter and Control Panel Integrated in Single Fabricated Box
	2		Make	-	Kirloskar Approved Vendor
	3		Material of Enclosure	-	Rittal Enclosure - Sheet Steel Fabricated Enclosure - CRCA Sheet
	4		Thickness of Enclosure	mm	Rittal Enclosure - (For Single Circuit Chillers) Enclosure - 1.5 mm Door - 2 mm Fabricated Enclosure - (For Dual Circuit Chillers) Load Bearing Member - 2 mm Non-Load Bearing Member - 1.6 mm
	5		Ingress Protection (IP)	-	IP54
	6		Painting Specification		
		i	Paint Type	-	RAL 7035
		ii	Standard	-	Coating as per KCPL Standards
	7		Mounting Arrangement	-	Mounted on Chiller
	8		Type of Starter	-	Star-Delta Starter
	9		Type of Isolation	-	Not Applicable
	10		Type of Protection	-	Fuses per Circuit
	11		Switchgear Make	-	Schneider
	12		Electrical and Control Cables	-	Power - PVC Insulated Single Core (Vtg. Grade 1.1 kV) Control- PVC Insulated Single Core, Multicore Cable (Vtg. Grade 1.1 kV) Signal- Shielded Cable
	13		Optional Features		
		i	Phase Indicating Lamps	-	Special-Optional
		ii	Hooter	-	Special-Optional
		iii	Energymeter	-	Special-Optional
O			Controller		
	1		Make	-	Refer "Make List" Sheet
	2		Transmitters	-	NA
	3		Oil Level Switch	-	Yes, Provided
	4		Oil Level Failure Trip	-	Yes, Provided
	5		LP Switch and Gauge	-	No, Controller Program will Take Care of Low Pressure
	6		HP Switch and Gauge	-	No, Controller Program will Take Care of High Pressure
	7		Chilled Water Flow Failure	-	Yes
	8		Cooling Water Flow Failure	-	Yes
	9		Reverse Rotor Protection	-	No
	10		High/Low Voltage Trip	-	Yes
	11		Low Current Trip (Current Based-Analog)	-	Yes
	12		High Current Trip (Current Based-Analog)	-	Yes
	13		Phase Failure/Reverse Phasing Trip	-	Yes
	14		Earth Fault Trip	-	No
	15		Communication Through RS232/RS485	-	RS485

Sr. No.			Description	UOM (Wherever Applicable)	Data (Common For All Models)
	16		Display of Microprocessor	-	Yes
	17		Type of Display	-	PGD0 Screen
	18		Remote Monitoring Facility	-	Yes
	19		Output to DCS	-	Applicable (Only if RS485 is Available)